



Oral Cavity, Palate & Salivary Glands

Anatomy GIT Block · MEDucation by Mattin Majid

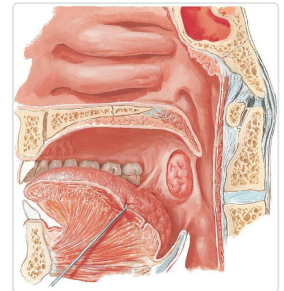
BEFORE YOU READ — ANSWER FROM MEMORY

1. Which soft-palate muscle is NOT supplied by the vagus, and which extrinsic tongue muscle is NOT supplied by the hypoglossal?
2. Where does the parotid duct open, and what does it pierce on the way?
3. Trace the parasympathetic pathway to the submandibular gland from the brainstem.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Oral cavity — parts and boundaries

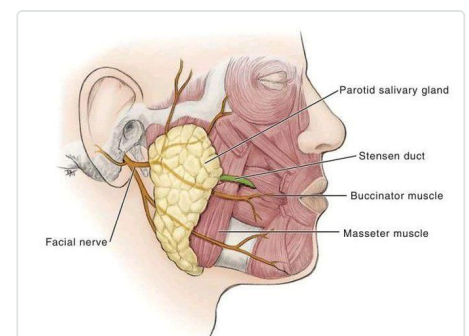
What	First part of the GIT: oral fissure → oropharynx
Vestibule	Space between lips/cheeks and teeth/gingiva (teeth and gingiva = its medial boundary)
Oral cavity proper	Space within the dental arches
Anterior · lateral	Lips · cheeks
Roof	Hard and soft palate
Floor	Tongue + mucosa over the mylohyoid region
Posterior	Opens into the oropharynx through the oropharyngeal isthmus (palatoglossal fold)



Sagittal section of the oral cavity.

	Lips	Cheeks
Main muscle	Orbicularis oris	Buccinator: presses cheek on teeth; helps chewing, sucking, blowing
Layers / role	Skin → subcutaneous tissue → muscle → glands → oral mucosa	Lateral wall of the oral cavity
Motor	Facial nerve (CN VII)	
Sensory	Upper lip: infraorbital (V2) · lower lip: mental (V3)	Buccal mucosa: buccal nerve (V3)

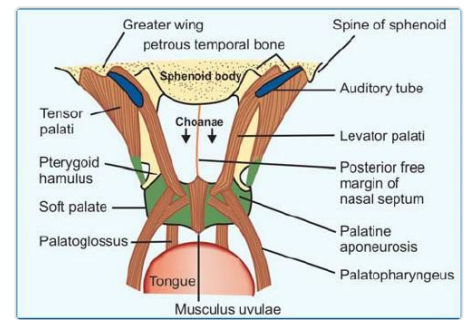
Parotid duct course: crosses the **masseter** → turns medially → **pierces buccinator** → opens into the vestibule **opposite the upper (maxillary) second molar**.



Parotid duct crossing masseter and piercing buccinator.

2 Palate

	Hard palate	Soft palate
Share	Anterior $\frac{2}{3}$	Posterior $\frac{1}{3}$
Nature	Bony, relatively immobile	Muscular, mobile; closes the nasopharynx in swallowing
Made of	Palatine processes of maxillae + horizontal plates of palatine bones	5 muscles (below) + palatine aponeurosis
Sensory	Greater palatine, lesser palatine, nasopalatine nerves	



Muscles of the soft palate.

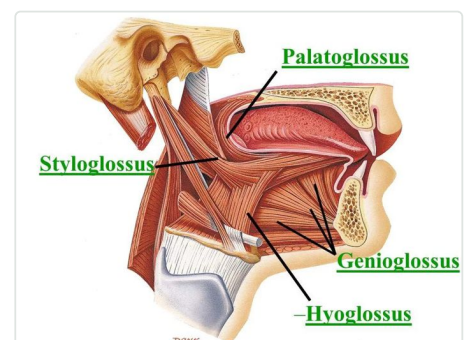
Soft palate muscle	Motor nerve
Tensor veli palatini	Mandibular nerve (V3) — the exception · forms most of the palatine aponeurosis
Levator veli palatini · palatoglossus · palatopharyngeus · musculus uvulae	Vagus (CN X) via the pharyngeal plexus

3 Floor of the mouth and teeth

Floor support	Mylohyoid muscles · contains tongue, sublingual glands, submandibular ducts
Duct openings	Submandibular duct → sublingual caruncle beside the lingual frenulum · sublingual gland → many ducts along the sublingual fold
Teeth	Adult: 32 · deciduous: 20 , with no premolars and no third molars · incisors cut, canines tear, premolars crush, molars grind
Arteries	External carotid branches: lingual (tongue, floor) · facial (lips, parts of cheek/palate) · maxillary (teeth, palate, deep structures) · rich supply helps healing but bleeds freely
General sensation	Mostly trigeminal (CN V)

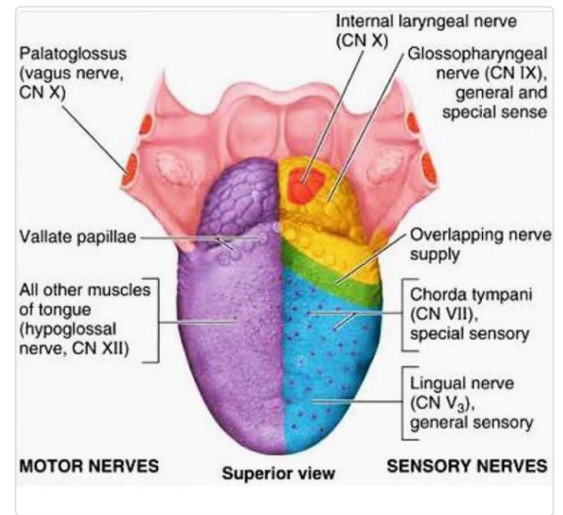
4 Tongue

Roles	Taste, speech, mastication, swallowing
Parts	Anterior $\frac{2}{3}$ = oral part · posterior $\frac{1}{3}$ = oropharyngeal part , separated by the sulcus terminalis
Foramen cecum	At the apex of the sulcus terminalis · remnant of the thyroglossal duct
Intrinsic muscles	Superior and inferior longitudinal, transverse, vertical → change shape
Extrinsic muscles	Genioglossus, hyoglossus, styloglossus, palatoglossus → move the tongue
Motor	Hypoglossal (CN XII) — except palatoglossus: vagus (CN X)



Extrinsic muscles of the tongue.

Region (from the slide figure)	General sensation	Taste
Anterior $\frac{2}{3}$	Lingual nerve (V3)	Chorda tympani (CN VII)
Posterior $\frac{1}{3}$ (incl. vallate papillae)	Glossopharyngeal (CN IX) — general and special	
Near the epiglottis	Internal laryngeal nerve (CN X)	
Lymph from	Drains mainly to	
Tip	Submental nodes	
Lateral anterior tongue	Submandibular nodes	
Central tongue	Bilateral deep cervical nodes	



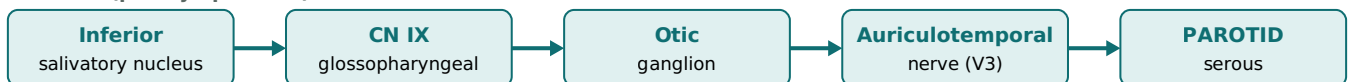
Motor and sensory nerves of the tongue.

5 Salivary glands — overview

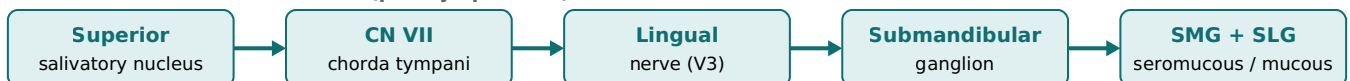
Function	Exocrine: saliva for mastication, swallowing, speech and starch digestion (salivary amylase)		
Major glands	Paired parotid, submandibular, sublingual → 90-95% of daily saliva		
Minor glands	Hundreds in the submucosa of lips, cheeks, palate and tongue		
	Parotid	Submandibular	Sublingual
Size	Largest	Middle	Smallest
Secretion	Serous	Mixed seromucous (mostly serous) · most resting saliva	Mainly mucous
Capsule	Tough parotid sheath (deep cervical fascia)	Deep cervical fascia over superficial part	No true capsule
Site	Retromandibular fossa	Submandibular triangle , U-shaped around the free edge of mylohyoid	Under floor-of-mouth mucosa, above mylohyoid
Duct	Stensen's → vestibule opposite upper 2nd molar	Wharton's → sublingual caruncle	Ducts of Rivinus (8-20) → sublingual fold
Parasympathetic	CN IX → otic ganglion	CN VII → chorda tympani → submandibular ganglion	

SECRETOMOTOR PATHWAYS — follow the nerve from brainstem to gland

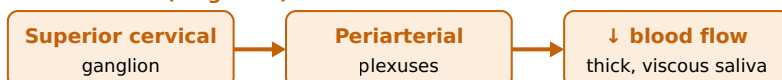
PAROTID (parasympathetic)



SUBMANDIBULAR + SUBLINGUAL (parasympathetic)

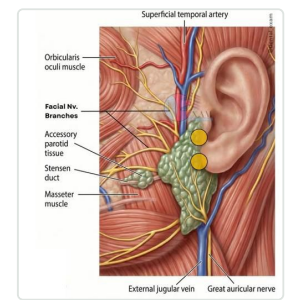


SYMPATHETIC (all glands)



6 Parotid gland

Position	Wedged between the mandibular ramus (front) and mastoid process + SCM (behind)
Anterior border	On the posterior border of the ramus · facial nerve branches and parotid duct emerge here
Posterior border	Overlaps the anterior border of sternocleidomastoid
Medial border	Points to the lateral pharyngeal wall ; wraps round the styloid process and carotid sheath structures

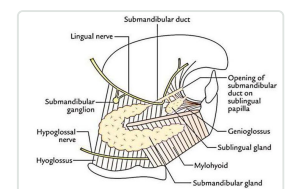


Superficial view of the parotid with the facial nerve branches and duct.

Surface	Relations
Superficial (lateral)	Skin, superficial fascia, platysma · pre-auricular (superficial parotid) nodes on the capsule · crossed by the great auricular nerve
Anteromedial	Posterior border of ramus, masseter and medial pterygoid insertions · facial nerve branches emerge from under its anterior border · Stensen's duct leaves its upper part and crosses the masseter
Posteromedial	SCM, posterior belly of digastric , styloid process and its muscles (styloglossus, stylohyoid, stylopharyngeus) · deep: ICA, IJV and CN IX-XII · above: tympanic plate (external acoustic meatus)

7 Submandibular and sublingual glands

Parts	Superficial part in the submandibular triangle (skin, platysma, deep fascia) · deep part curls around the free border of mylohyoid into the floor of the mouth
Borders	Mandible body above · digastric bellies below
Bed	Mylohyoid, hyoglossus, middle constrictor
Facial artery	Loops over its superior and posterior borders
Nerves	Lingual and hypoglossal pass deep · submandibular ganglion hangs from the lingual nerve
Wharton's duct	From the deep part → forward → crosses lateral to the lingual nerve (per slides) → sublingual caruncle at the frenulum base
Sublingual gland	Raises the sublingual fold · medial: genioglossus + lingual nerve · lateral: sublingual fossa of mandible · anterior ducts may join Wharton's duct as Bartholin's duct



Submandibular duct, lingual nerve and submandibular ganglion.

All salivary glands	
Arteries	External carotid branches: transverse facial, posterior auricular, facial, lingual
Drainage	Parotid → retromandibular / jugular chains via parotid nodes · submandibular + sublingual → submandibular and deep cervical nodes

P Must-remember pearls

- Oral cavity = vestibule (lips/cheeks ↔ teeth) + oral cavity proper (inside the arches); opens back through the **palatoglossal fold**.
- Lips: orbicularis oris (CN VII) · upper lip **V2 infraorbital**, lower lip **V3 mental** · buccal mucosa **V3 buccal**.
- Parotid duct: over masseter → **pierces buccinator** → opens opposite the **upper 2nd molar**.
- Hard palate $\frac{2}{3}$ (maxillary palatine processes + palatine horizontal plates) · soft palate $\frac{1}{3}$, closes the nasopharynx.
- Soft palate: all **CN X** except **tensor veli palatini (V3)**, which forms the palatine aponeurosis.
- Tongue: all **CN XII** except **palatoglossus (CN X)** · intrinsic = shape, extrinsic = movement.
- Sulcus terminalis divides anterior $\frac{2}{3}$ / posterior $\frac{1}{3}$ · foramen cecum = **thyroglossal duct** remnant.
- Tongue lymph: tip → submental · lateral → submandibular · central → **bilateral** deep cervical.
- Teeth: 32 adult · 20 deciduous (**no premolars, no third molars**).

- Parotid = largest, serous, retromandibular fossa · submandibular = seromucous, **most resting saliva** · sublingual = smallest, mucous, **no capsule**.
- Parotid: **IX → otic ganglion → auriculotemporal** · SMG/SLG: **VII → chorda tympani → lingual → submandibular ganglion** · sympathetic: thick, viscous saliva.
- Submandibular: facial artery loops over it; ganglion hangs from the lingual nerve; Wharton's duct → sublingual caruncle.

Q Test yourself

1. After a dental block a patient cannot feel his lower lip, but he can still move it normally. Which nerve has been anaesthetised?

- | | |
|----------------------------|--------------------------|
| A. Infraorbital nerve (V2) | B. Facial nerve (CN VII) |
| C. Buccal nerve (V3) | D. Mental nerve (V3) |
| E. Hypoglossal nerve | |

2. A patient with a lesion of the right hypoglossal nerve can still use one extrinsic tongue muscle normally. Which one?

- | | |
|--------------------------|-----------------|
| A. Palatoglossus | B. Genioglossus |
| C. Hyoglossus | D. Styloglossus |
| E. Superior longitudinal | |

3. A stone blocks the duct that opens on the sublingual caruncle beside the lingual frenulum. Which gland will swell?

- | | |
|--------------------|------------------------|
| A. Parotid | B. Sublingual only |
| C. Submandibular | D. Minor labial glands |
| E. Palatine glands | |

4. A surgeon cuts the chorda tympani during middle-ear surgery. Which glands lose their secretomotor supply?

- | | |
|--|---------------------------|
| A. Parotid only | B. Lacrimal gland |
| C. Minor palatine glands only | D. All three major glands |
| E. Submandibular and sublingual glands | |

5. During parotidectomy, which structures must be identified as they emerge from the anterior border of the gland?

- | | |
|--|---|
| A. Branches of the facial nerve and the parotid duct | B. The lingual nerve and Wharton's duct |
| C. The internal carotid artery and CN IX | D. The great auricular nerve and platysma |
| E. The submandibular ganglion | |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 D — Sensation of the lower lip is from the mental nerve (V3); lip movement is by the facial nerve, which is intact.

2 A — All tongue muscles are supplied by CN XII except palatoglossus, which is supplied by the vagus.

3 C — Wharton's (submandibular) duct opens on the sublingual caruncle; sublingual ducts open along the sublingual fold.

4 E — Submandibular and sublingual secretomotor fibres run CN VII → chorda tympani → lingual nerve → submandibular ganglion. The parotid uses CN IX via the otic ganglion.

5 A — Branches of the facial nerve and Stensen's duct emerge from or cross the anterior border of the parotid.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Anatomy GIT Block · MEDucation by Mattin Majid · GIT Anatomy LG 1 slides, 2026-2027
Study alongside the lecture, not instead of it.